Quaia Redshift Anisotropy & FR Cosmology Fits ,Äî Updated Results (0V2)

EXECUTIVE SUMMARY

This document summarises the updated analysis of large-scale anisotropy in the Quaia G<20.0 quasar catalogue and its connection to FR (Full Relativity) cosmological fits using DESI DR2 BAO and Pantheon+SH0ES data. It extends the previous 0V1 report by:

,Ä¢ Including the full command and processing pipeline used to obtain the Quaia G<20 selection.

,Ä¢ Adding the selection,Äìcorrected redshift-dipole fits and redshift-binned dipoles.

,Ä¢ Describing the residual quadrupole in the mean-redshift map and its significance versus randomised mocks, including a |b|>30¬∞ mask variant.

,Ä¢ Summarising new FR cosmology fits to DESI DR2 BAO alone, with different c(z) coupling choices and Noether-link settings.

The key result is that the Quaia mean-redshift field exhibits a statistically extreme large-scale quadrupole (even after removing a best-fit dipole and masking the Galactic plane), while BAO-only FR fits point to non-standard combinations of H0, Œ©m and an additional acceleration sector with moderate œá¬≤ per degree of freedom.

1. ACRONYMS, SYMBOLS, AND BASIC DEFINITIONS

GC: Galactic Centre (l = 0¬∞, b = 0¬∞).

anti-GC: Galactic anti-centre (l = 180¬∞, b = 0¬∞).

CMB dipole: Direction of the CMB temperature dipole (l ,âà 264¬∞, b ,âà 48¬∞).

Quaia: Gaia,ÄìunWISE quasar catalogue (version 0.1.0, DOI 10.5281/zenodo.8060755).

DESI DR2 BAO: Baryon Acoustic Oscillation distance measurements from DESI Data Release

2.FR: Full Relativity cosmological framework with an additional acceleration term A_acc and exponent n_acc.

H0: Hubble constant at z = 0 (km s,Åª¬π Mpc,Åª¬π).

Œ©m, Œ©Œõ, Œ©k: Present-day matter, dark-energy and curvature density parameters.

A_acc, n_acc: Amplitude and exponent of the extra acceleration sector in FR.

Œ≥c, ŒµM: FR parameters controlling effective c(z) behaviour and luminosity/magnitude evolution, linked via the soft Noether constraint.

w0, wa: CPL dark-energy equation-of-state parameters.

l, b: Galactic longitude and latitude.

Œîz: Residual mean redshift after subtracting the fitted dipole model.

C,Ñì, D,Ñì: Angular power spectrum and its rescaled form D,Ñì = ,Ñì(,Ñì+1)C,Ñì / 2œÄ.

D2: Quadrupole power at ,Ñì = 2, expressed as D2 = 2¬∑3 C2 / 2œÄ.

œÉ-significance: (measured ,àí mean(mock)) / std(mock) based on random mocks.

2. DATASETS AND PRE-PROCESSING

2.1 Quaia G<20.0 Sample

The Quaia catalogue (DOI 10.5281/zenodo.8060755) provides a Gaia,ÄìunWISE quasar sample with improved photometric redshifts. For this analysis, the G<20.0 subset is used, with the following main columns: source_id, redshift_quaia, ra, dec, l, b, and photometric magnitudes.

Key conversion steps:
1. Download the full Quaia suite via download_zenodo_quaia.py.
2. Extract the G<20.0 science catalogue (quaia_G20.0.fits).
3. Convert to a compact CSV via quaia_to_csv.py, selecting RA, DEC, and redshift_quaia:
    Output: quaia_G20p0_basic.csv (columns: RA, DEC, Z).
4. Optionally add Galactic coordinates (L_GAL, B_GAL) via qso_add_galactic_coords.py.

Basic statistics for the Quaia G<20.0 sample:
,Ä¢ N ,âà 755,850 objects.
,Ä¢ RA ,àà [0¬∞, 360¬∞), DEC ,àà [,àí90¬∞, 90¬∞].
,Ä¢ Z ,àà [0.084, 4.537] with median z ,âà 1.45,Äì1.50.

2.2 Random Catalogue and Selection Function
The Quaia release includes random_G20.0_10x.fits and selection_function_NSIDE64_G20.0.fits.

The random catalogue approximates the angular and magnitude selection; the selection map encodes the relative completeness per HEALPix pixel at nside = 64.

These are used to:
,Ä¢ Build randomised mock realisations with the same angular selection as the data.
,Ä¢ Define an analysis mask by requiring sel ,â• f √ó max(sel) with f ,àà [0.2, 0.5], typically f = 0.3.

2.3 DESI DR2 BAO and Pantheon+SH0ES

The FR cosmology runs use:
,Ä¢ DESI_DR2_BAO.csv and DESI_DR2_BAO_cov.txt (13 BAO data points).
,Ä¢ Optionally Pantheon+SH0ES SN Ia (Pantheon+SH0ES.dat with the full STAT+SYS covariance) for joint FR fits. One such joint run was attempted but halted when emcee encountered infinite parameter values for aggressive link-strength settings.

3. QUAIA COUNTS AND REDSHIFT DIPOLES

3.1 Number-Count Dipole
The initial analysis computed a simple number-count dipole in Galactic coordinates by summing unit vectors for all quasars.

Global dipole (all 0.08 ,â≤ z ,â≤ 4.5):
,Ä¢ N = 755,850, amplitude ,âà 2.83 √ó 10,Åª¬≤.
,Ä¢ Direction (l, b) ,âà (231.1¬∞, 41.8¬∞).

Redshift-binned number-count dipoles (z bins: [0,1], [1,2], [2,4]) show broadly similar orientations with modest variation in amplitude, reinforcing a mild number-count anisotropy.

3.2 Mean-Redshift Dipole (No Selection Function)
A HEALPix map of the mean redshift ,ü®z,ü© was constructed at nside = 64, requiring a minimum per-pixel count.

Full redshift range, no selection function:
,Ä¢ zÃÑ ,âà 1.53, |D| ,âà 4.18 √ó 10,Åª¬≥, fractional amplitude ,âà 2.74 √ó 10,Åª¬≥.
,Ä¢ (l, b) ,âà (191.4¬∞, 8.8¬∞).

3.3 Selection-Function,ÄìCorrected Redshift Dipole
Including the Quaia selection map at nside = 64, with sel ,â• 0.3 √ó max(sel) and ,â•10 objects per pixel:

,Ä¢ zÃÑ ,âà 1.5135, |D| ,âà 9.16 √ó 10,Åª¬≥, fractional amplitude ,âà 6.05 √ó 10,Åª¬≥.
,Ä¢ Direction (l, b) ,âà (180.5¬∞, 13.7¬∞).

Varying sel-min-frac between 0.2 and 0.5 leaves the dipole direction broadly stable (l ,âà 175¬∞,Äì185¬∞).

3.4 Redshift-Binned Dipoles with Selection Function
Selection-function,Äìcorrected redshift dipoles in three z bins:

,Ä¢ 0.0 ,â§ z < 1.0:
zÃÑ ,âà 0.667, |D| ,âà 3.0 √ó 10,Åª¬≥, frac amplitude ,âà 4.5 √ó 10,Åª¬≥,
(l, b) ,âà (172¬∞, ,àí10¬∞).

,Ä¢ 1.0 ,â§ z < 2.0:
zÃÑ ,âà 1.48, |D| ,âà 2.6 √ó 10,Åª¬≥, frac amplitude ,âà 1.7 √ó 10,Åª¬≥,
(l, b) ,âà (166¬∞, ,àí26¬∞).

,Ä¢ 2.0 ,â§ z < 4.0:
zÃÑ ,âà 2.50, |D| ,âà 1.07 √ó 10,Åª¬≤, frac amplitude ,âà 4.3 √ó 10,Åª¬≥,
(l, b) ,âà (102¬∞, 33¬∞).

The low- and mid-z bins maintain an anti-centre‚Äìlike direction; the high-z bin rotates significantly.

4. RESIDUAL QUADRUPOLE IN MEAN-z AND RANDOM MOCKS

4.1 Dipole Removal and Residual Map

Using quaia_dipole_residual_map.py, the best-fit redshift dipole was subtracted to form a residual map Œîz(nÃÇ) = ‚ü®z‚ü©(nÃÇ) ‚àí z_dipole_model(nÃÇ), with hp.UNSEEN in unobserved pixels.

The angular power spectrum C‚Ñì of Œîz up to ‚Ñìmax = 10 gives, for the full masked sky:
‚Ä¢ C‚ÇÇ ‚âà 4.86 √ó 10‚Åª¬≥, D‚ÇÇ ‚âà 4.64 √ó 10‚Åª¬≥.

4.2 Quadrupole-Only Map and Axis

A pure quadrupole map was constructed via map2alm up to ‚Ñì=2, zeroing ‚Ñì=0,1 and reconstructing via alm2map.

Max |Œîz| in the quadrupole map:
‚Ä¢ (l, b) ‚âà (185.6¬∞, 84.2¬∞) (near the Galactic north pole).

4.3 Random Mocks and Significance (No |b| Cut)

quaia_random_mocks_zdipole_quad.py built 50 random realisations using random_G20.0_10x and the same selection mask.

Mock ensemble:
‚Ä¢ dipole_frac (mock): mean ‚âà 2.07 √ó 10‚Åª¬≥, std ‚âà 8.89 √ó 10‚Åª‚Å¥.
‚Ä¢ D‚ÇÇ_residual (mock): mean ‚âà 4.19 √ó 10‚Åª‚Å∂, std ‚âà 2.67 √ó 10‚Åª‚Å∂.

Real map (same mask, no |b| cut):
‚Ä¢ dipole_frac_real ‚âà 6.05 √ó 10‚Åª¬≥ ‚Üí ‚âà 4.5œÉ above the mock mean.
‚Ä¢ D‚ÇÇ_real ‚âà 4.64 √ó 10‚Åª¬≥ ‚Üí formally ‚âà 1.7√ó10¬≥œÉ above the mock mean.

This huge D‚ÇÇ significance shows that the residual quadrupole is not explained by the random catalogue alone.

4.4 |b|>30¬∞ Cut and Revised Quadrupole Significance

Imposing an additional |b| ‚â• 30¬∞ cut:

Real map:
‚Ä¢ C‚ÇÇ_cut ‚âà 2.00 √ó 10‚Åª‚Å¥, D‚ÇÇ_cut ‚âà 1.91 √ó 10‚Åª‚Å¥.

50 mocks with identical |b|-cut:
‚Ä¢ D‚ÇÇ_residual (mock, b‚â•30¬∞): mean ‚âà 2.13 √ó 10‚Åª‚Å∂, std ‚âà 1.40 √ó 10‚Åª‚Å∂.
‚Üí z_D2_b30 ‚âà 135œÉ.

Thus even outside the Galactic plane, the residual quadrupole is extraordinarily large.

5. UPDATED FR COSMOLOGY FITS WITH DESI DR2 BAO

5.1 FR + DESI DR2 BAO, c_of_z, Soft Link

Command (schematic):
PLamb_Test_10V6c_plus.py --out plamb_runs/desi_tests/FR_w0_softlink_sigma0p6_desionly --model FR --flat --de-model cpl --bao DESI_DR2_BAO.csv --bao-cov DESI_DR2_BAO_cov.txt --rd 147.09 --bao-c-mode c_of_z --c-model log1pz --link-soft --kappa-em-gc 1.0 --link-sigma 0.60 --noether combo_soft --noether-strength 1.0 ...

Active parameters: {H0, Œ©m, A_acc, n_acc, Œ≥c, ŒµM, w0} with wa fixed = 0.

Best-fit (approx):
,Ä¢ H0 ,âà 50.9 km s,Åª¬π Mpc,Åª¬π.
,Ä¢ Œ©m ,âà 0.134, Œ©Œõ ,âà 0.866 (flat FR).
,Ä¢ A_acc ,âà 0.71, n_acc ,âà 1.20.
,Ä¢ Œ≥c ,âà ,àí0.30, ŒµM ,âà 0.49.
,Ä¢ w_eff ,âà ,àí0.60.

Goodness-of-fit:
,Ä¢ œá¬≤_BAO ,âà 6.73 for dof ,âà 6 ,Üí œá¬≤/dof ,âà 1.12.
,Ä¢ Link + Noether penalty ,âà 4.0 in œá¬≤.

The chain is mildly under-sampled (steps_post/œÑ_max ,âà 25.6) but indicates that DESI BAO alone prefer a low-H0, moderate-Œ©m FR solution with non-trivial extra acceleration.

5.2 FR + DESI DR2 BAO, c0 BAO Mode, Stronger Noether Link

Command (schematic):
PLamb_Test_10V6c_plus.py --out plamb_runs/desi_tests/FR_CPL_softlink_c0_desionly_v2 --model FR --flat --de-model cpl --bao DESI_DR2_BAO.csv --bao-cov DESI_DR2_BAO_cov.txt --rd 147.09 --bao-c-mode c0 --c-model log1pz --link-soft --kappa-em-gc 1.0 --link-sigma 0.25 --noether combo_soft --noether-strength 1.5 ...

Best-fit summary:
,Ä¢ H0 ,âà 58.9 km s,Åª¬π Mpc,Åª¬π.
,Ä¢ Œ©m ,âà 0.40, Œ©Œõ ,âà 0.60.
,Ä¢ A_acc ,âà 0.34, n_acc ,âà 0.54.
,Ä¢ Œ≥c ,âà 0.0024, ŒµM ,âà 0.223.
,Ä¢ w_eff ,âà ,àí0.82.

Goodness-of-fit:
,Ä¢ œá¬≤_BAO ,âà 9.18 for dof ,âà 5 ,Üí œá¬≤/dof ,âà 1.84.
,Ä¢ Link residual ŒµM ,àí Œ∫Œ≥c ,âà 0.22 ,Üí œá¬≤_link ,âà 0.78; Noether ,âà 1.04; total

extra ‚âà 1.82.

This run pushes towards higher H0 and Œ©m with a more ŒõCDM-like effective w but a worse œá¬≤/dof than the c_of_z run.

5.3 Joint SN+BAO FR Run and Instability
A joint FR run including Pantheon+SH0ES SN + DESI DR2 BAO in a full-covariance setup, with strong link-soft and Noether-strength, halted when emcee reported infinite parameter values, signalling numerical or prior issues. This suggests that the combined data plus strong symmetry-breaking priors tightly constrain the allowed FR parameter region and may require re-tuned priors or reparametrisation.

6. RELATION TO PREVIOUS BAO/HHT RESULTS
Previous HHT_Locking and BAO-residual work identified characteristic BAO scales and potential quasi-periodic residuals in SN Ia data. The current Quaia analysis adds a complementary sky-map perspective:

‚Ä¢ Instead of SN residuals, it examines the mean-redshift field of a deep quasar survey.
‚Ä¢ The extremely large quadrupole in ‚ü®z‚ü©, after dipole removal and |b|>30¬∞ masking, hints at either a major systematic or a real large-scale anisotropy.
‚Ä¢ If physical, such a quadrupole could be linked to the same underlying acceleration or c(z) physics probed in FR/HHT analyses, potentially tied to large-scale structure or symmetry-breaking effects.

7. SCRIPTS AND PIPELINES USED
Main scripts:
‚Ä¢ download_zenodo_quaia.py ‚Äì download Quaia FITS and auxiliary files.
‚Ä¢ quaia_to_csv.py ‚Äì convert quaia_G20.0.fits to RA/DEC/Z CSV.
‚Ä¢ qso_add_galactic_coords.py ‚Äì add Galactic (l, b) to CSV.
‚Ä¢ qso_dipole_catalog.py ‚Äì compute number-count dipoles and count maps.
‚Ä¢ quaia_redshift_dipole.py ‚Äì build mean-z maps and fit redshift dipoles.
‚Ä¢ quaia_redshift_dipole_sel.py ‚Äì as above, but with selection map masking.
‚Ä¢ quaia_dipole_residual_map.py ‚Äì produce dipole model and residual mean-z maps; compute C‚Ñì.
‚Ä¢ quaia_random_mocks_zdipole_quad.py ‚Äì generate random mocks and dipole/quadrupole statistics.
‚Ä¢ AngularSeperation.py ‚Äì compute angular separations between axes (Quaia, GC, CMB, etc.).
‚Ä¢ PLamb_Test_10V6c_plus.py ‚Äì FR cosmology engine for SN+BAO with flexible c(z), Œõ(z), and Noether options.

8. CONCLUSIONS AND NEXT STEPS
1. The Quaia G<20.0 catalogue exhibits a robust mean-redshift dipole aligned roughly with the Galactic anti-centre but at positive latitude, with fractional amplitude ‚âà 6 √ó 10‚Åª¬≥ under selection masking.
2. After dipole removal, the residual mean-z field shows an extraordinarily large quadrupole.

D,ÇÇ is orders of magnitude above random-mock expectations, even for |b|>30¬∞, suggesting either a strong unmodelled systematic or a genuine cosmic anisotropy.
3. FR BAO-only fits using DESI DR2 favour non-standard combinations of H0, Œ©m, and A_acc with reasonable œá¬≤/dof, and are sensitive to the assumed c(z) model and Noether-link strength.
4. Fully joint FR fits with SN+BAO and strong symmetry-breaking priors are numerically delicate, indicating highly constrained or unstable directions in parameter space.
5. Combining Quaia anisotropy, BAO distance constraints, and SN/HHT residual structure offers a promising multi-probe framework to test FR-style extensions and cosmic anisotropy hypotheses.

Planned next steps:
,Ä¢ Extend the mock analysis (more mocks, alternative selection-function treatments).
,Ä¢ Explore redshift-dependent masks and cross-correlation of Quaia Œîz with large-scale structure tracers.
,Ä¢ Refine FR joint fits (SN+BAO+Planck) with more robust priors and lower-dimensional parameterisations.
,Ä¢ Compare orientations of Quaia quadrupole/dipole axes with BAO- and SN-derived anisotropy axes from previous HHT work.